HEALTH360

CLINICAL & HEALTHCARE RISK ANALYSIS

DETAILED PROJECT REPORT

An exploratory data-visualization project using Tableau

Project reference	Details
Project title	Health360 Clinical & Healthcare Risk Analysis
Dataset size stated in project brief	5,000 patient/person records
Dataset fields stated in project brief	17 columns; exact field names to be verified against source file
Analytics platform	Tableau
Visual deliverables	Nine worksheets/visualizations, a dashboard, and a Tableau story
Repository	https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git
Project website	https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/

Demo video: Open the Health360 project demonstration on Google Drive

Prepared as an academic project report. Replace any verification notes with evidence from the final workbook, dataset, and published views before formal submission.

Document Control and Contents

Purpose of this report

This report documents the purpose, analytical workflow, design decisions, implementation approach, validation plan, limitations, and future scope of the Health360 project. It expands the supplied two-page structure into a report suitable for review and finalization. Known project details are stated directly; information not supplied in the source brief—such as the exact column names, chart titles, and numerical findings—is identified as requiring confirmation rather than invented.

Document item	Value
Document title	Health360 Clinical & Healthcare Risk Analysis — Detailed Project Report
Version	1.0 — expanded report draft
Prepared for	Academic project documentation and review
Primary tool	Tableau
Repository	https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git
Website	https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/
Demonstration video	https://drive.google.com/file/d/1EVdFMnmq4gDKbIZzebHt5hB2Y4B3zz_J/view?usp=drive_link

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3. Abstract and Executive Summary

Abstract

Health360 Clinical & Healthcare Risk Analysis is an educational, exploratory data-visualization project intended to make a healthcare-related dataset easier to inspect and compare. The project brief describes a dataset containing 5,000 patient/person records and 17 columns. Tableau is used to develop nine worksheets or visualizations, combine selected views into a dashboard, and organize an accompanying story titled around Patient Risk & Lifestyle Insights. A project website and a source-code repository provide supporting access to the project deliverables.

The purpose of the project is to support structured exploration of patterns represented in the supplied data, not to diagnose patients, predict clinical outcomes, recommend treatment, or replace professional judgement. This report describes a reproducible workflow for data inspection, preparation, metric definition, visualization design, publication, and testing. Exact dataset fields, chart-level outputs, and quantitative findings must be taken from the original data and final Tableau workbook before the report is treated as a verified results document.

Executive summary

- **Problem addressed:** Flat tables can make it difficult to compare multiple attributes and identify patterns across groups. Interactive visualizations can make those comparisons more accessible.
- **Approach:** Review the source data, define analytical questions, prepare fields, create worksheet-level views, assemble a dashboard, and sequence key views into a Tableau story.
- **Deliverables described in the project brief:** Nine worksheets/visualizations, a dashboard, a story, a GitHub repository, and a project website.
- **Expected value:** Improve data exploration, provide a consistent visual interface, and help a reader formulate questions for further investigation.
- **Boundary:** Findings are descriptive unless a separate, appropriate statistical design supports stronger conclusions. No clinical validity is claimed.

Verification note

The supplied project structure does not include the actual dataset header, chart screenshots, or numerical chart outputs. Therefore, this report does not fabricate risk percentages, rankings, correlations, or medical conclusions. Add the verified values and screenshots from the published workbook in the final submission.

4. Introduction, Context, and Problem Statement

4.1 Background

Healthcare-related datasets can contain multiple attributes describing people, lifestyle factors, health indicators, or other observations. When those attributes are viewed only as rows and columns, relationships and differences may be difficult to notice quickly. A well-designed dashboard provides a consistent visual surface for comparing measures, groups, and distributions while preserving the ability to inspect the underlying data.

Data visualization does not make data inherently correct. Its value depends on the quality of the source, the definitions of metrics, the choices made during aggregation, and the way uncertainty is communicated. Health360 is therefore framed as an exploratory analytics exercise: it organizes data for inspection and communicates patterns that should be checked against the source data.

4.2 Problem statement

The project addresses the practical challenge of exploring a multi-field healthcare-related dataset in a clear and navigable way. Users need a structured way to review selected measures, compare groups, and move from an overall view to more focused questions. A dashboard and story can reduce the effort required to navigate separate charts, provided that each visualization has a clear purpose and accurate labels.

4.3 Rationale for Tableau

- Tableau supports visual exploration through worksheets, dashboards, filters, and interactive views.
- A worksheet-oriented workflow allows individual visualizations to be developed and checked before they are assembled.
- Dashboards can place related views together so that users can compare them without switching between separate files.
- A story can sequence views into a guided narrative while leaving the underlying observations open to further exploration.
- Published views and a companion website can make the result easier to demonstrate and review.

4.4 Intended audience

The intended audience is an academic evaluator, project teammate, or general reader interested in learning how a healthcare-related dataset can be explored visually. The project is not presented as a validated clinical tool for patients or clinicians. Any real-world clinical use would require appropriate governance, validation, privacy controls, and specialist review beyond the scope of this academic deliverable.

4.5 Report organization

The following sections describe objectives and requirements, the dataset and data preparation, methodology, visual design, implementation, testing, interpretation boundaries, and recommendations for future development.

5. Objectives, Scope, Users, and Requirements

5.1 Project objectives

1. Organize the supplied healthcare-related data into a form suitable for systematic visual exploration.
2. Develop nine worksheet-level visualizations, as stated in the project brief, with each view serving a defined analytical question.
3. Combine related views into a dashboard that supports comparison and navigation.
4. Create a Tableau story around Patient Risk & Lifestyle Insights to guide the reader through the selected views.
5. Publish supporting project assets through the project website and repository, and provide a demonstration video for review.
6. Document assumptions, data-quality checks, limitations, and future improvements.

5.2 Scope included

- Exploratory and descriptive analysis of the supplied dataset.
- Data-field review, basic cleaning and validation where required, and documentation of transformations.
- Worksheet, dashboard, and story development in Tableau.
- Publication checks for the views and website, including link and browser testing.
- Academic documentation and a demonstration walkthrough.

5.3 Scope excluded

- Clinical diagnosis, treatment recommendations, or patient-level medical advice.
- Claims that observed associations prove causation.
- A validated predictive model unless separately developed and evaluated with an appropriate methodology.
- Any use or publication of identifiable patient information without explicit authorization and appropriate safeguards.
- Claims of performance or findings that have not been verified against the dataset and published workbook.

5.4 Functional requirements

ID	Requirement	Acceptance evidence
FR-01	Load and inspect the approved source dataset.	Source file and field list recorded.
FR-02	Build nine worksheet-level views.	Workbook contains nine identified worksheets/views.
FR-03	Provide a consolidated dashboard.	Dashboard loads and displays intended views.
FR-04	Provide a story for guided exploration.	Story sequence and captions reviewed.
FR-05	Publish and link project assets.	Repository, website, and demo links open.
FR-06	Document data and interpretation limits.	Report includes limitations and verification notes.

6. Dataset Profile and Data Governance

6.1 Known dataset profile

The supplied project brief identifies the dataset as containing 5,000 patient/person records and 17 columns. It does not provide the source publisher, source URL, license, exact field names, units, data dictionary, collection period, or the distinction between synthetic and real-world records. Those items must be confirmed from the original data file or its accompanying documentation.

Profile item	Known status / action
Record count	5,000 records stated in the project brief; reconcile with the actual loaded row count.
Column count	17 columns stated in the project brief; compare with the source header.
Field names and types	Not present in the supplied report structure; transcribe from source.
Source and publisher	To be recorded from the original dataset page or provider.
License and permitted use	Verify before redistribution or publication.
Missing values and duplicates	Measure from the actual file; do not assume none exist.
Sensitive data status	Determine whether fields can identify a person or reveal sensitive information.

6.2 Data dictionary

A data dictionary should be included as an appendix or companion file. For each of the 17 columns, record the exact field name, plain-language definition, data type, unit or allowed values, missing-value convention, and any transformation applied. The dictionary should distinguish original source fields from calculated fields created in Tableau.

Field	Definition	Type / unit	Validation rule
Copy exact source field 1	Use source documentation	Confirm	Define after inspection
Copy exact source field 2	Use source documentation	Confirm	Define after inspection
… continue through all 17 fields …	Do not infer definitions	Confirm	Document applicable checks

6.3 Data governance and privacy

- Use only data that the project team is authorized to access and process.
- Avoid publishing direct identifiers, small groups that could enable re-identification, or row-level details that are not necessary for the demonstration.
- Keep a copy of the original source file and perform transformations on a working copy so that changes can be traced.
- Record the source, access date, license, and any restrictions on redistribution.
- If the data are synthetic or public, state that accurately; do not imply that they represent actual clinical records unless the source establishes this.

Data protection principle

The report should describe the dataset conservatively until its provenance and privacy status are confirmed. The label “patient/person records” in the project brief is not, by itself, evidence that the records are real patient records.

7. Data Preparation and Quality Assurance

7.1 Preparation workflow

Data preparation should be carried out before interpreting the charts. The steps below form a practical checklist; report only the actions that were actually executed and retain a cleaning log to support them.

1. **Preserve the source:** retain an unchanged original dataset and record the filename, source, and access date.
2. **Inspect structure:** verify row and column counts, field names, data types, and whether each row represents one intended observation.
3. **Check missingness:** count blank, null, and placeholder values per field; decide whether to retain, exclude, or recode them based on the field meaning.
4. **Check duplicates:** define what constitutes a duplicate record before removing any rows. Similar-looking records are not automatically duplicates.
5. **Validate ranges and categories:** identify impossible values, inconsistent spelling, mixed units, and category labels that refer to the same concept.
6. **Review outliers:** flag unusual values for investigation rather than deleting them automatically.
7. **Document transformations:** record any parsing, grouping, calculated fields, exclusions, or standardization performed.
8. **Reconcile counts:** compare the number of rows before and after each transformation and explain any difference.

7.2 Quality checks and evidence

Check	What to calculate or inspect	Evidence to retain
Structure	Actual rows and columns versus stated 5,000 / 17.	Profile screenshot or log.
Completeness	Null/blank counts by field and percentage.	Field-level missingness table.
Uniqueness	Duplicate count using a defined key or rule.	Duplicate check result.
Validity	Data types, allowable values, plausible ranges.	Rules and exceptions list.
Consistency	Category spelling, units, date formats.	Before/after examples.
Reconciliation	Counts and key totals before/after cleaning.	Calculation log.

7.3 Handling missing and unusual values

Missing data should be treated according to its meaning and prevalence. A missing value may indicate an unrecorded observation rather than a zero. Replacing missing values with zero can materially distort a measure, so it should only be done when zero is a valid and documented meaning. If records or categories are excluded, record the reason and disclose how exclusions affect the population represented by each chart.

7.4 Reproducibility

A reproducible project should preserve the original dataset, maintain a short transformation log, document calculated fields, and identify the workbook version used for publication. If the source data change, refresh the workbook and repeat the checks before replacing the published views.

8. Analytical Methodology and Metric Design

8.1 Analytical approach

The appropriate baseline approach for this project is descriptive exploratory data analysis. It summarizes what is represented in the supplied dataset and helps users investigate differences across categories or measures. It does not establish why a pattern occurred, and it should not be described as predictive or causal analysis unless additional methods and evidence support those claims.

8.2 Define each analytical question

Before building a worksheet, write one sentence specifying the question it answers. Examples of question types—subject to the actual available fields—include the distribution of a numeric measure, differences across categories, how two variables vary together, or how a selected subgroup compares with the full dataset. The final report must use the exact questions answered by the actual nine worksheets.

8.3 Metric specification

Metric element	Documentation needed
Name	A concise, unambiguous label matching the chart.
Definition	What the metric measures and which source field(s) it uses.
Aggregation	Count, distinct count, sum, average, median, rate, or another defined operation.
Denominator	For percentages or rates, state the population used as the denominator.
Grouping	The dimension or category used to split the data.
Filters	Default filter state and any exclusions applied.
Formatting	Units, rounding, date granularity, and treatment of missing values.
Validation	How the displayed value was reconciled to the source or a separate calculation.

8.4 Calculations and interpretation

For counts, the report should clarify whether the measure is a row count or a distinct-person count. For percentages, document numerator and denominator and confirm that both use the same filter context. For averages, inspect the distribution and consider whether a median or distribution chart is also needed. For any comparison, check that groups are defined consistently and have adequate sample sizes for the intended display.

8.5 Correlation is not causation

If two variables appear to move together, describe the observation as an association within the dataset. Confounding variables, selection effects, missingness, and data collection practices may explain or influence the pattern. A visual comparison alone does not show that one factor caused another or that an observed relationship will generalize to a broader population.

8.6 Validation protocol

- Recreate selected totals independently using the source data or a separate calculation.
- Check that filters affect all intended views consistently.
- Verify that axis scales, labels, units, and category order match the data.
- Test the effect of excluding missing values and note material changes.
- Record the workbook date/version and the filter state for every reported finding.

9. Visualization and Dashboard Design

9.1 Worksheet-level design

The project brief reports nine worksheets or visualizations. Each worksheet should have a single primary analytical purpose and should make the measure, grouping, and population clear. Chart types must be selected from the actual question and data type rather than for visual variety alone.

Design concern	Recommended practice
Chart title	Use a descriptive title that states the subject and relevant population or period.
Axes and units	Label axes; include units and date granularity where applicable.
Comparison	Use a consistent scale when comparing related groups or panels.
Sorting	Sort categories intentionally and explain ranking criteria when needed.
Color	Use a restrained palette; do not rely on color alone to communicate meaning.
Missing data	Show or disclose missing values when they materially affect interpretation.
Tooltips	Include useful context such as metric definition, group, and count where appropriate.
Accessibility	Use readable type, adequate contrast, and descriptive text.

9.2 Dashboard composition

The dashboard should establish a visual hierarchy: an informative title, a small set of summary indicators only if they are meaningful and validated, the primary analytical views, and controls that are clearly labeled. Related charts should be positioned close together. Filter controls should be easy to locate, and the user should be able to recognize when a filter changes the population being summarized.

9.3 Interaction and navigation

- Ensure filter names correspond to the field and values they control.
- Set sensible default states and provide a way to return to an overall view.
- Use dashboard actions only when the relationship between source and target views is understandable.
- Check that tooltips do not obscure important information or imply unsupported conclusions.
- Test the dashboard at the expected presentation size and in the browser where it will be demonstrated.

9.4 Tableau story: Patient Risk & Lifestyle Insights

The project brief states that a Tableau story was part of the workflow and associates it with Patient Risk & Lifestyle Insights. A useful story sequence should move from context and dataset orientation to the most relevant descriptive views, then close with a balanced summary and limitations. Each story point should explain what the viewer should look at without presenting an association as a cause or a descriptive measure as a diagnosis.

9.5 Screenshots to insert

For the final submission, insert actual screenshots of the published dashboard, the story opening, representative worksheet views, and any important filter interaction. Caption each image with the view name, the metric shown, and the filter state used. Do not replace real screenshots with illustrative mock-ups.

10. Implementation, Publishing, and Website

10.1 Implementation overview

The described workflow uses Tableau for worksheet development, dashboard assembly, and story presentation. The repository and project website act as supporting delivery points. The final report should identify the actual workbook filename, publication date, and the URLs that were tested at submission time.

Component	Role	Verification
Source dataset	Input for exploration and visualization.	Confirm provenance, schema, license, and row count.
Tableau worksheets	Individual visual views; nine are stated in the brief.	Confirm names and formulas in the workbook.
Dashboard	Consolidated interface for comparison and interaction.	Open published dashboard and test controls.
Tableau story	Guided sequence around Patient Risk & Lifestyle Insights.	Open story and check order, captions, and navigation.
GitHub repository	Project source and supporting files.	Confirm files, README, and current access.
Project website	Landing point for the project and embedded or linked assets.	Test on desktop and mobile browser.
Demo video	Walkthrough for review and assessment.	Confirm permission and playback access.

10.2 Project links

Repository: https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git

Project website: https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/

Demonstration video: Watch the Health360 project demo on Google Drive

10.3 Publishing checks

1. Open each published Tableau view directly and verify that it loads without an authentication or permission error.
2. Check that the published workbook reflects the intended latest version and that captions and titles are correct.
3. Open the project website in a clean browser session and test every dashboard/story link or embed.
4. Check the layout at common desktop and mobile widths; confirm that the content remains readable.
5. Open the demo link in a separate browser session to confirm that the reviewer has viewing permission.
6. Record the final URLs and the date on which they were tested.

10.4 Reproducibility and maintenance

Store project documentation alongside the source files where appropriate. Use clear filenames and a versioning convention. If data, calculations, or published views are updated, revise the report and re-run the validation checklist so that the narrative remains aligned with the live deliverables.

11. Testing, Validation, and Evaluation

11.1 Testing principles

A visual analytics project needs both data-level checks and interface-level tests. A chart that renders correctly can still contain an incorrect aggregation; likewise, a correct calculation can be presented in a confusing or inaccessible interface. The team should record test cases, expected outcomes, actual outcomes, and any fixes.

Test ID	Test	Expected outcome	Status to record
T-01	Load source dataset	Schema and row count reconcile to source.	Pass / Fail / Not run
T-02	Validate calculated fields	Independent calculation agrees within defined rounding.	Pass / Fail / Not run
T-03	Check missing values	Treatment is consistent with documented rules.	Pass / Fail / Not run
T-04	Test dashboard filters	Only intended views change; totals remain coherent.	Pass / Fail / Not run
T-05	Check labels and units	Titles, axes, legends, and units are accurate.	Pass / Fail / Not run
T-06	Open story navigation	Story points and captions display in the intended order.	Pass / Fail / Not run
T-07	Open published views	Dashboard and story load from their public URLs.	Pass / Fail / Not run
T-08	Website browser test	Links/embeds work and layout remains usable.	Pass / Fail / Not run
T-09	Demo access	Reviewer can open and play the video.	Pass / Fail / Not run
T-10	Privacy/accessibility review	No unauthorized sensitive data; text and contrast are usable.	Pass / Fail / Not run

11.2 Calculation reconciliation

Select a sample of key totals, percentages, and grouped values from the dashboard. Recalculate them independently from the source data, applying the same exclusions and filter conditions. Record the comparison and explain any mismatch. A discrepancy may arise from aggregation settings, null handling, distinct-count logic, duplicated records, or filter context.

11.3 Usability and accessibility review

- Can a first-time viewer identify the subject of the dashboard quickly?
- Are metric labels and units understandable without verbal explanation?
- Are filter controls and reset behavior discoverable?
- Does the page remain readable when embedded in the project website?
- Is information conveyed by labels or symbols as well as color?
- Are links descriptive and usable with keyboard navigation where supported?

11.4 Test reporting

Do not mark a test as passed unless it was actually executed. The final report should replace the status placeholders with the test results, date, tester, and short notes on failures and fixes. Any test not performed should remain marked “Not run” or be removed with a clear explanation.

12. Results Interpretation, Limitations, and Risks

12.1 How to report results

The supplied project structure does not contain the final chart outputs or verified numeric findings. As a result, this report intentionally avoids asserting specific risk levels, lifestyle relationships, rankings, or percentages. In the final version, every result should be tied to an actual chart and should identify the population, metric, denominator where relevant, filter state, and source view.

Recommended format for each finding

Element	What to write
Finding	One concise sentence describing the pattern visible in the chart.
Population	The records included, with relevant exclusions and filters.
Metric	Exact measure, aggregation, units, and denominator if applicable.
Evidence	Worksheet/dashboard title and screenshot or page reference.
Qualification	Missingness, sample size, alternative explanations, or other caveats.
Implication	What question the pattern suggests for further analysis—not a medical instruction.

12.2 Interpretation boundaries

- **Descriptive scope:** A pattern describes the dataset as represented; it may not generalize to other populations.
- **Association:** A relationship between variables does not establish causality.
- **Data quality:** Missing values, selection bias, inconsistent definitions, or measurement error can affect results.
- **Sample size:** Small groups may produce unstable or misleading comparisons and should be labeled carefully.
- **Clinical meaning:** A dashboard metric is not a validated diagnostic threshold or a treatment recommendation.
- **Data provenance:** Conclusions depend on the source, collection process, definitions, and permitted use of the dataset.

12.3 Project risks and mitigations

Risk	Potential impact	Mitigation
Unverified source or fields	Incorrect definitions or claims.	Verify provenance and maintain a data dictionary.
Incorrect aggregation	Misleading KPI or comparison.	Document formulas and reconcile independently.
Overinterpretation	Clinical or causal claims unsupported by data.	Use cautious language and state limitations.
Public link failure	Evaluator cannot inspect deliverables.	Test URLs and permissions before submission.
Privacy exposure	Disclosure of sensitive information.	Review fields and publish only authorized content.
Outdated published views	Report and live dashboard disagree.	Record versions and repeat checks after updates.

13. Conclusion, Future Work, References, and Appendices

13.1 Conclusion

Health360 demonstrates an exploratory analytics workflow built around a healthcare-related dataset and Tableau. The project brief identifies 5,000 records, 17 columns, nine worksheets or visualizations, a dashboard, and a Tableau story associated with Patient Risk & Lifestyle Insights. The companion repository, website, and demonstration video provide ways to review the project beyond the report itself.

The project's quality depends on more than the visual presentation. Reliable interpretation requires verified data provenance, clear field definitions, documented cleaning decisions, transparent metric calculations, and tests of both the workbook and its published interface. The project should be described as educational and exploratory; it is not a clinically validated diagnostic, prognostic, or treatment system.

13.2 Future work

- Complete a source-backed data dictionary for all 17 columns and document provenance and licensing.
- Add a reproducible data-quality report covering missingness, duplicates, valid ranges, and row-count reconciliation.
- Add verified screenshots and chart-specific findings to the report, including metric definitions and filter states.
- Improve accessibility through stronger contrast, readable text, clear labels, and descriptive captions.
- Document the refresh procedure and maintain a versioned record of data and workbook changes.
- If a future predictive or clinical use is proposed, establish a separate governance and validation plan with qualified domain experts.

13.3 References and project resources

The final submission should cite the actual dataset source and any external documentation used. The following are project links supplied in the project brief or in the current request; confirm that each remains accessible before submission.

Project repository: GitHub — Health360 Healthcare Risk

Project website: Health360 project website

Demo video: Google Drive demonstration video

Method reference: Tableau documentation for worksheets, dashboards, stories, filters, and calculated fields should be cited using the specific pages consulted during development.

13.4 Final submission checklist

- Replace verification notes with the exact dataset source, field list, and observed quality checks.
- Insert actual screenshots of the dashboard, story, and representative worksheets.
- Add at least three to five verified findings supported by chart-level evidence, if appropriate to the actual analysis.
- Complete the testing table with real outcomes and dates.
- Verify repository, website, Tableau views, and Google Drive demo permissions.
- Add author, course, institution, guide/evaluator, and submission date if required by the institution.

Project demonstration video

Click here to watch the Health360 project demo

End of report